

Reducing Cybersickness for 2D VR Spectators Using a Gaze-Based Stabilized Third-Person View

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ABSTRACT

While motion sickness in VR has been widely studied for HMD (head-mounted display) wearers, the experience of external spectators watching VR content on 2D monitors has received relatively little attention. This paper proposes a gaze-based stabilized third-person view that adjusts the spectator camera according to the HMD gaze behavior of the HMD wearer. Through a repeated measures experiment using the Simulator Sickness Questionnaire (SSQ), we show that the proposed view reduces cybersickness experienced by spectators watching VR content on 2D monitors, compared to conventional first-person and third-person spectator views.

Index Terms: Virtual Reality, Spectator Cybersickness, Third-Person Viewpoint, Gaze-Based Camera Control, Simulator Sickness Questionnaire.

1 INTRODUCTION

Virtual reality (VR) enables immersive interaction within three-dimensional environments and has rapidly expanded across various application domains. While extensive research has addressed cybersickness experienced by HMD wearers, the viewing experience of external spectators who watch VR content on 2D monitors has received relatively little attention.

In practice, VR content is commonly shared by directly transmitting the first-person view of the HMD wearer. As a result, subtle head movements and rapid viewpoint rotations are directly reflected in the video, often leading to visual inconsistency and increased motion sickness for spectators [1]. Although several approaches have been proposed to alleviate spectator discomfort, such as silhouette-based observation or free-viewpoint spectating systems, their applicability is limited due to restricted viewpoint representation or additional system complexity [2, 3].

Motivated by prior findings that third-person viewpoints induce less motion sickness than first-person views for VR users [4], this study investigates whether a similar effect can be observed in 2D spectator environments. However, conventional third-person views still involve frequent viewpoint rotations and transitions, which can continue to induce discomfort. To address this limitation, we propose a gaze-based stabilized third-person viewing method that uses the HMD wearer's gaze behavior as a criterion for viewpoint control. Through experimental evaluation, we demonstrate that the proposed approach effectively reduces motion sickness experienced by spectators. Beyond the reported reduction in motion sickness, this study addresses an underexplored scenario by focusing on cybersickness experienced by spectators watching VR content on standard 2D monitors.

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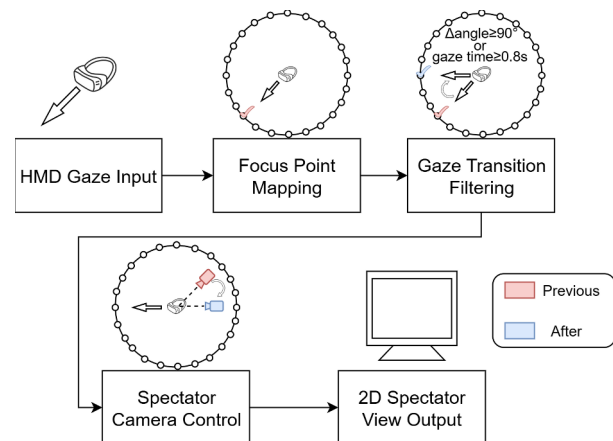


Figure 1: Overview of the proposed gaze-responsive third-person spectator view pipeline.

2 METHODS

2.1 Gaze-Responsive Third-Person Viewpoint Control

Figure 1 illustrates the overall pipeline of the proposed gaze-responsive third-person spectator view. In this work, the wearer's "gaze" is operationalized as the HMD forward viewing direction derived from the first-person camera pose (i.e., HMD orientation), rather than eye-tracking. We place 24 focus points evenly spaced at 15° intervals on a circle located far from the avatar in world space, serving as discrete look-at direction anchors. At runtime, the system computes the gaze direction from the local forward vector of the first-person camera (HMD pose) and selects the closest focus point. The spectator camera maintains a conventional third-person back-view configuration with a fixed height and a fixed distance behind the avatar, while its orientation is controlled to look toward the selected focus point so that the avatar remains stably centered in the 2D view.

To prevent unnecessary viewpoint oscillations, viewpoint transitions are gated by two conditions: a transition is triggered only when the gaze remains near a focus point for at least 0.8 s, or when the gaze direction changes by more than 90° relative to the current active focus point. When a viewpoint transition is triggered, the camera does not instantly rotate to the new look-at target in a single frame; instead, it updates its rotation in a time-based manner so that the viewpoint changes smoothly with an approximately constant angular velocity. Once the remaining orientation error (and position error, if adjusted) becomes sufficiently small, a snap condition is applied to precisely align the view, ensuring continuity without unnatural shaking.

This design stabilizes the 2D spectator view in two ways. First, discretizing the look-at target into finite focus points and applying temporal gating suppresses short-term jitter by preventing rapid switching between neighboring targets. Second, the angular threshold and smooth rotation reduce excessive camera rotation by avoiding frequent small updates and easing large changes.

2.2 Experimental Design and Procedure

To evaluate the effectiveness of the proposed approach, participants watched three videos rendered from the same FPS VR scene play-back, differing only in viewpoint: first-person, conventional third-person, and gaze-responsive third-person views. All factors other than viewpoint were controlled by recording the same gameplay segment across conditions. The VR gameplay was performed using a Meta Quest 3 HMD, and the resulting spectator videos were presented to participants on a 2D monitor. Participants were seated and instructed to watch the videos attentively without interacting with the system. The videos were presented full-screen under the same viewing setup for all participants.

After watching each video, participants reported their motion sickness using the Simulator Sickness Questionnaire (SSQ) [5], a widely used symptom checklist that quantifies nausea, oculomotor, and disorientation related discomfort after visually induced motion exposure. Although SSQ was originally developed for simulator/VR exposure, we use it here as a standardized symptom inventory to compare motion sickness related symptoms across the three spectator viewing conditions in a controlled, within-subjects design. While “cybersickness” typically refers to discomfort experienced by HMD wearers, the symptoms reported by 2D spectators are closer to visually induced motion sickness (VIMS). Therefore, SSQ scores are interpreted as preliminary symptom-level comparisons rather than definitive clinical diagnoses. To minimize carry-over effects (e.g., fatigue or residual discomfort), participants were given sufficient rest between videos before proceeding to the next condition. All participants watched the three videos in the same fixed order. A total of 17 participants took part in the experiment (12 men, 5 women; ages 20–29: $n = 10$, 30–39: $n = 7$), and all participants experienced all three viewing conditions.

3 RESULTS AND ANALYSIS

As a result of the experiment, a consistent decreasing trend in motion sickness scores was observed across all SSQ subscales (Nausea, Oculomotor, and Disorientation) as well as the total score, from the first-person view to the conventional third-person view and further to the proposed gaze-responsive third-person view (Table 1, Figure 2). The first-person condition produced the highest level of motion sickness, while the conventional third-person view showed a partial reduction.

The proposed gaze-responsive third-person view yielded the lowest scores across all subscales and the total score, indicating the most effective reduction of motion sickness for spectators. In particular, pronounced reductions were observed in the Nausea and Disorientation subscales (Table 1, Figure 2), which can be attributed to the suppression of unnecessary rotations and abrupt viewpoint changes during camera transitions.

These findings suggest that switching to a third-person perspective can help reduce motion sickness, but that a more stable third-person viewpoint further decreases spectator discomfort by smoothing viewpoint transitions using HMD-based gaze information. Statistical analysis was conducted using one-way repeated-measures ANOVA across the three viewpoint conditions. Since Mauchly’s test indicated violations of sphericity, Greenhouse–Geisser corrections were applied. The analysis revealed a significant main effect of viewpoint on the SSQ total score, $F(1.192, 19.070) = 8.227$, $p = .007$, $\eta_p^2 = .340$, and Bonferroni-corrected post-hoc comparisons showed that the proposed view significantly reduced total SSQ scores compared to both the first-person ($p = .013$) and conventional third-person ($p < .001$) views. Standard deviation (SD) is reported in Table 1 to reflect inter-participant variability, whereas standard error (SE) is used in Figure 2 to highlight differences between condition means.

Table 1: SSQ scores (mean $\pm$ standard deviation) for different spectator views

SSQ Subscale	1st-person	Conventional 3rd-person	Gaze-responsive 3rd-person
Nausea	40.40 $\pm$ 32.97	30.30 $\pm$ 26.61	19.64 $\pm$ 22.49
Oculomotor	60.64 $\pm$ 47.56	47.26 $\pm$ 36.11	28.98 $\pm$ 35.07
Disorientation	74.51 $\pm$ 84.09	51.59 $\pm$ 62.59	30.30 $\pm$ 58.49
SSQ Total	65.78 $\pm$ 57.87	49.06 $\pm$ 43.41	30.14 $\pm$ 40.27

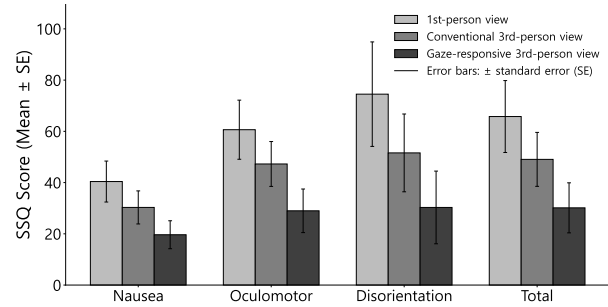


Figure 2: SSQ scores (mean $\pm$ standard error) for different spectator viewpoints.

4 CONCLUSION

This study proposes a gaze-responsive third-person spectator viewpoint that utilizes gaze information derived from HMD orientation and verifies its effectiveness in reducing motion sickness among 2D monitor–based spectators compared to conventional first-person and general third-person views. Experimental results show that the proposed method yields the lowest SSQ scores across all subscales, with particularly notable reductions in disorientation. The results further suggest that third-person viewpoints mitigate spectator discomfort and that stabilizing viewpoint transitions—beyond merely changing viewpoint position—plays a key role in additional reduction. Limitations include the small sample size ($n = 17$) and the lack of measures for immersion or preference, and fixed-order viewing may have introduced order and individual-susceptibility effects; thus, the results should be interpreted as preliminary. Future work will conduct larger-scale studies and examine the trade-off between sickness reduction and overall viewing experience.

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